Curent alternativ

$x(t) = X_{max} \sin(\omega t + \varphi)$

- X_{max} → amplitudinea / val. de vârf / val. max
- ω → pulsația / frecvența unghiulară
- φ → faza inițială

$\omega = 2\pi f = \frac{2\pi}{T}$ → perioada

f → frecvența

$X_{max} = X\sqrt{2}$ → val. efectivă

$x_1(t) = X_1\sqrt{2}\sin(\omega t + \varphi_1)$

$x_2(t) = X_2\sqrt{2}\sin(\omega t + \varphi_2)$

$\Rightarrow \varphi = \varphi_1 - \varphi_2$ DEFAZAJ

Transformarea din timp în complex

$x(t) = X\sqrt{2}\sin(\omega t + \varphi) \Rightarrow \underline{X} = x \cdot e^{j\varphi}$

$= x(\cos\varphi + j\sin\varphi)$

ex $i(t) = 2\sqrt{2}\sin(\omega t - \frac{\pi}{2})$

$\underline{i} = 2 \cdot e^{j(-\frac{\pi}{2})} = 2\left(\cos(-\frac{\pi}{2}) + j\sin(-\frac{\pi}{2})\right) = 2(0 + j(-1))$

$\underline{i} = -2j$

ex $e(t) = 4\sin(\omega t + \frac{\pi}{4})$

$\underline{e} = \frac{4}{\sqrt{2}} e^{j\frac{\pi}{4}} = \frac{4}{\sqrt{2}}\left(\cos(\frac{\pi}{4}) + j\sin(\frac{\pi}{4})\right) = \frac{4}{\sqrt{2}} \cdot \frac{\sqrt{2}}{2}(1 + j)$

$\underline{e} = 2(1 + j)$

Transformare din complex în timp

$$\underline{X} = a + bj$$

$$x(t) = \sqrt{a^2+b^2} \cdot \sqrt{2} \sin\left(\omega t + \operatorname{arctg} \frac{b}{a}\right) \text{ pt. } a \geq 0$$

$$x(t) = \sqrt{a^2+b^2} \cdot \sqrt{2} \sin\left(\omega t + \operatorname{arctg} \frac{b}{a} + \pi\right) \text{ pt. } a < 0$$

ex $\underline{I} = 5 \Rightarrow i(t) = \sqrt{5^2+0^2} \cdot \sqrt{2} \sin\left(\omega t + \operatorname{arctg} \frac{0}{5}\right)$

$$i(t) = 5\sqrt{2} \sin(\omega t)$$

ex $\underline{I} = -5 \Rightarrow i(t) = \sqrt{5^2+0^2} \cdot \sqrt{2} \sin\left(\omega t + \operatorname{arctg} \frac{0}{5} + \pi\right)$

$$i(t) = 5\sqrt{2} \sin(\omega t + \pi)$$

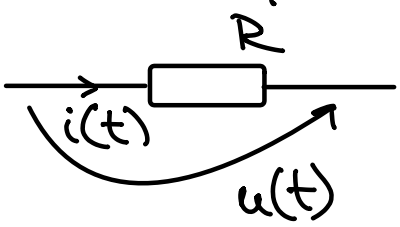
ex $\underline{U} = 1 - j \Rightarrow u(t) = \sqrt{1^2+(-1)^2} \cdot \sqrt{2} \cdot \sin\left(\omega t + \operatorname{arctg} \frac{-1}{1}\right)$

$$u(t) = 2 \sin\left(\omega t - \frac{\pi}{4}\right)$$

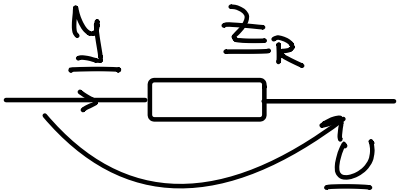
$$u(t) = 2 \sin\left(\omega t + \frac{7\pi}{4}\right)$$

Elemente de circuit

→ Rezistența



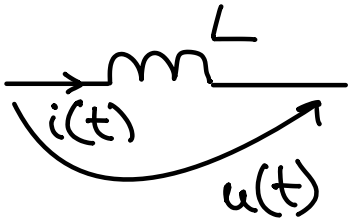
$$u(t) = R\,i(t)$$

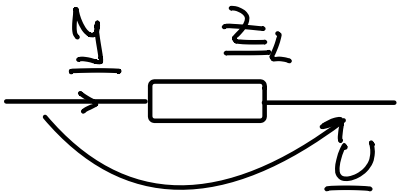


$$\underline{U} = \underline{Z}_R \cdot \underline{I}$$

$\underline{Z}_R = R$ impedanță rezistivă

→ Bobina





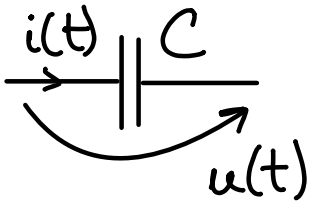
L = inductivitate

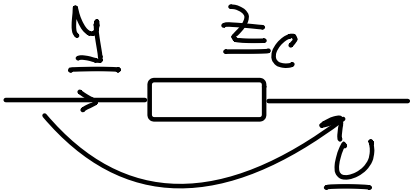
$$u(t) = L \cdot \frac{di(t)}{dt}$$

$\underline{Z}_L = j\omega L$ impedanță inductivă

$X_L = \omega L$ reactanță inductivă

→ Condensator





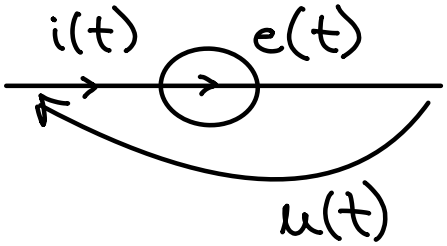
C = capacitate

$$i(t) = C \cdot \frac{du(t)}{dt}$$

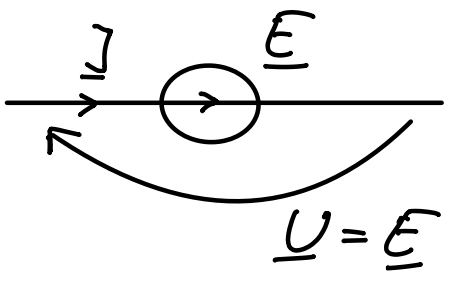
$$\underline{Z}_C = \frac{1}{j\omega C} = \frac{-j}{\omega C}$$ impedanță capacitivă

$X_C = \frac{1}{\omega C}$ reactanță capacitivă

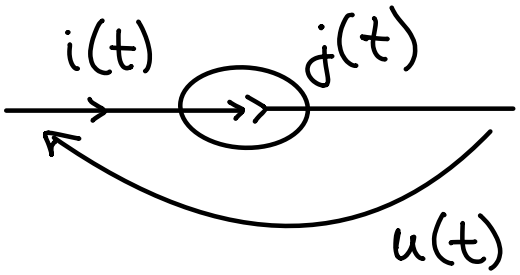
→ Sursă de tensiune

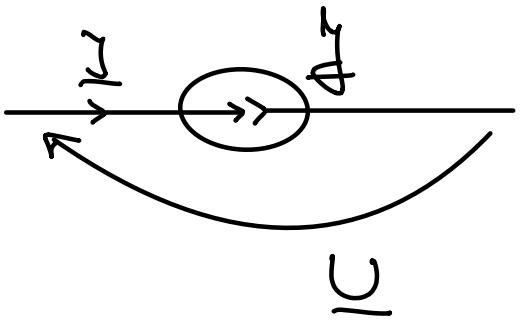


$u(t) = e(t)$



→ Sursă de curent





$$i(t) = j(t)$$

Bilanțul puterilor

P - puterea activă [W]

S - puterea reactivă [VAR]

Q - puterea aparentă [VA]

$$\underline{S}_c = \underline{S}_g \qquad \underline{S}_c = P_c + jQ_c$$

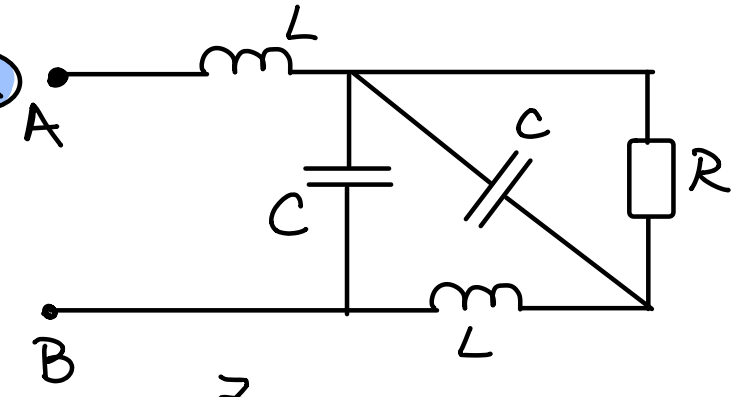
$$P_c = \sum Re\{\underline{Z}_k\} \cdot |\underline{Y}_k|^2$$

$$Q_c = \sum Im\{\underline{Z}_k\} \cdot |\underline{Y}_k|^2$$

$$\underline{S}_g = \sum (\underline{E}_k \cdot \underline{Y}_k^* + \underline{U}_k \cdot \underline{y}_k^*)$$

$$\left.\begin{aligned} P_g &= Re\{\underline{S}_g\} = P_c \\ Q_g &= Im\{\underline{S}_g\} = Q_c \end{aligned}\right| \Rightarrow \underline{S}_g = \underline{S}_c$$

ex



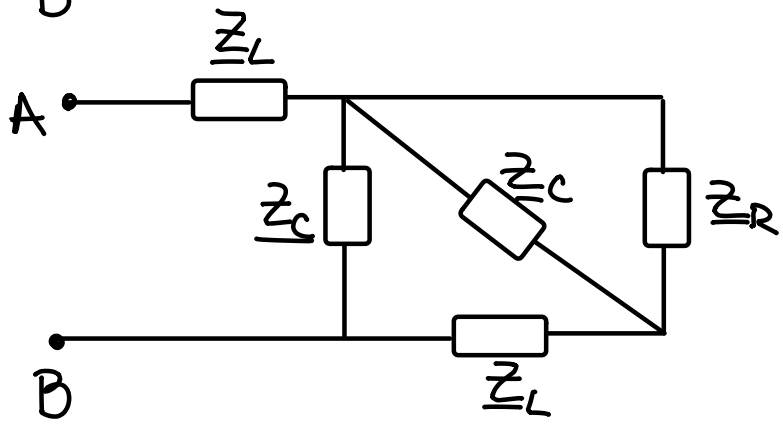
$R = 2\,\Omega$

$L = \frac{20}{\pi}\,mH$

$C = \frac{5}{\pi}\,mF$

$f = 50\,Hz$

$Z_{AB} = ?$



$$\omega = 2\pi f = 2\pi \cdot 50 = 100\pi$$

$$\underline{Z}_R = R = 2$$

$$\underline{Z}_L = j\omega L = j \cdot 100\pi \cdot \frac{20}{\pi} \cdot 10^{-3} = 2j$$

$$\underline{Z}_C = \frac{-j}{\omega C} = \frac{-j}{100\pi \cdot \frac{5}{\pi} \cdot 10^{-3}} = \frac{-10}{5}j = -2j$$

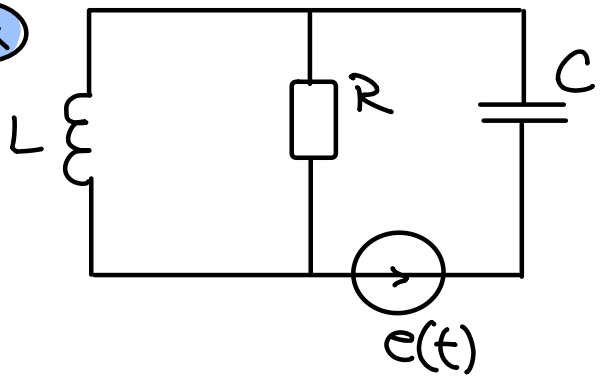
$$\underline{Z}_{P1} = \frac{\underline{Z}_C \cdot \underline{Z}_R}{\underline{Z}_C + \underline{Z}_R} = \frac{2 \cdot (-2j)}{2 - 2j} = \frac{-2j(1+j)}{1+1} = -j + 1 = 1 - j$$

$$\underline{Z}_{S1} = \underline{Z}_{P1} + \underline{Z}_L = 1 + j$$

$$\underline{Z}_{P2} = \frac{\underline{Z}_C \cdot \underline{Z}_{S1}}{\underline{Z}_C + \underline{Z}_{S1}} = \frac{-2j(1+j)}{1-j} = \frac{-2j(1+j)^2}{2j} = -j(1+2j-1) = 2$$

$$\underline{Z}_{AB} = \underline{Z}_{P2} + \underline{Z}_L = 2 + 2j$$

ex



$R = 5\,\Omega$

$L = \frac{50}{\pi}\,mH$

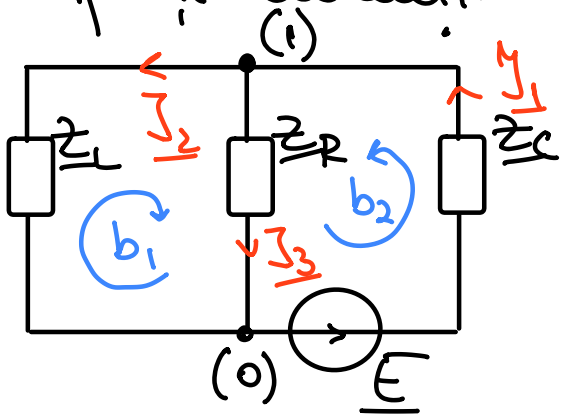
$C = \frac{2}{\pi}\,mF$

$f = 50\,Hz$

$e(t) = 120\sqrt{2}\sin(\omega t - \pi)\,[V]$

---

a) Aflați curenții



$\underline{Z}_R = R = 5\,\Omega$

$\omega = 2\pi f = 2\pi \cdot 50 = 100\pi \text{ rad/s}$

$\underline{Z}_L = j\omega L = j \cdot 100\pi \cdot \frac{50}{\pi} \cdot 10^{-3} = 5j$

$\underline{Z}_C = \frac{-j}{\omega C} = \frac{-j}{100\pi \cdot \frac{2}{\pi} \cdot 10^{-3}} = -5j$

10^{-1}

$\underline{E} = 120 e^{j(-\pi)} = 120\left(\cos(-\pi) + j\sin(-\pi)\right) = 120(-1+0) = -120$

MK

TK1: (1) $\underline{J}_1 - \underline{J}_2 - \underline{J}_3 = 0$

TK2 (b_1): $\underline{Z}_R \underline{J}_3 - \underline{Z}_L \underline{J}_2 = 0$

(b_2): $\underline{Z}_C \underline{J}_1 + \underline{Z}_R \underline{J}_3 - \underline{E} = 0$

MCC

$\underline{I}_1 = \underline{I}_{b2}$

$\underline{I}_2 = -\underline{I}_{b1}$

$\underline{I}_3 = \underline{I}_{b1} + \underline{I}_{b2}$

$\underline{Z}_R(\underline{I}_{b1} + \underline{I}_{b2}) - \underline{Z}_L(-\underline{I}_{b1}) = 0$

$\underline{Z}_C \underline{I}_{b2} - \underline{Z}_R(\underline{I}_{b1} + \underline{I}_{b2}) - \underline{E} = 0$

MPN

$\underline{V}_0 = 0$

$\underline{I}_1 = \dfrac{\underline{V}_0 - \underline{V}_1 + \underline{E}}{\underline{Z}_C}$

$\underline{I}_2 = \dfrac{\underline{V}_1 - \underline{V}_0}{\underline{Z}_L}$

$\underline{I}_3 = \dfrac{\underline{V}_1 - \underline{V}_0}{\underline{Z}_R}$

$$\frac{\underline{V}_0 - \underline{V}_1 + \underline{E}}{\underline{Z}_C} - \frac{\underline{V}_1 - \underline{V}_0}{\underline{Z}_L} - \frac{\underline{V}_1 - \underline{V}_0}{\underline{Z}_R} = 0$$

$$\frac{0 - \underline{V}_1 - 120}{-5j} - \frac{\underline{V}_1}{5j} - \frac{\underline{V}_1}{5} = 0 \quad | \cdot 5j$$

$$\cancel{\underline{V}_1} + 120 - \cancel{\underline{V}_1} - \underline{V}_1 j = 0$$

$$j\underline{V}_1 = 120 \Rightarrow \underline{V}_1 = \frac{120}{j} = -120j \text{ V}$$

$$\underline{I}_1 = \frac{120j - 120}{5j} = \frac{1}{j} \cdot (24 - 24j) = -24j - 24 = -24 - 24j$$

$$\underline{I}_2 = \frac{-120j}{5j} = -24$$

$$\underline{I}_3 = \frac{-120j}{5} = -24j$$

$$i_1(t) = \frac{48}{\sqrt{2}}\sqrt{2}\,\sin\left(\omega t + \frac{5\pi}{4}\right)$$

$$i_2(t) = 24\sqrt{2}\,\sin\left(\omega t + \pi\right)$$

$$i_3(t) = 24\sqrt{2}\,\sin\left(\omega t + \frac{\pi}{2}\right)$$

b) Faceți bilanțul puterilor

$$P_c = \mathrm{Re}\{\underline{Z}_C\}|\underline{I}_1|^2 + \mathrm{Re}\{\underline{Z}_L\}|\underline{I}_2|^2 + \mathrm{Re}\{\underline{Z}_R\}|\underline{I}_3|^2$$

$$= 0\cdot(24\sqrt{2})^2 + 0\cdot 24^2 + 5\cdot 24^2 = 2880\,W$$

$$Q_c = \mathrm{Im}\{\underline{Z}_C\}|\underline{I}_1|^2 + \mathrm{Im}\{\underline{Z}_L\}|\underline{I}_2|^2 + \mathrm{Im}\{\underline{Z}_R\}|\underline{I}_3|^2$$

$$= -5(24\sqrt{2})^2 + 5\cdot 24^2 + 0\cdot 24^2 = -2880\ VAR$$

$$\underline{S}_g = \underline{E}\cdot\underline{I}_1^* = -120\cdot(-24+24j) = 2880 - 2880j$$

$$P_g = \mathrm{Re}\{\underline{S}_g\} = 2880 = P_c$$

$$Q_g = \mathrm{Im}\{\underline{S}_g\} = -2880 = Q_c$$

$$\Rightarrow \underline{S}_c = \underline{S}_g$$